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Brain Research 1048 (2005) 48 – 58

BRAIN RESEARCH

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Research report

Inhibition from the plantar nerve to soleus muscle during the stance phase of walking

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Accepted 15 April 2005
Available online 25 May 2005

Abstract

The purpose of this study is to investigate the effect and the circuit from the branch of tibial (plantar) nerve to soleus muscle and its modulation during walking in humans.

Stimulation of the plantar nerve produced short latency inhibition of soleus EMG activity and the H-reflex in humans. The threshold of afferent fibers was lower than that of motor fibers. This inhibition did not converge to disynaptic reciprocal Ia inhibition nor did inhibition from the cutaneous nerve of the big toe, but to Ib inhibition from the medial gastrocnemius nerve. The inhibitory pathway from the plantar nerve therefore is considered to include Ib inhibitory interneurones.

Modulation of the inhibition was investigated during walking. Less EMG depression after plantar nerve stimulation occurred in the stance phase of walking than for tonic or dynamic plantar flexion at similar background EMG activity level. The inhibition of the soleus H-reflex after plantar nerve stimulation was also decreased during the stance phase. For investigating the influence of load on the inhibition from the plantar nerve, more EMG depression occurred in the stance phase with body unloading. Similar findings were observed in Ib inhibition from the medial gastrocnemius nerve, but not in disynaptic reciprocal Ia inhibition to soleus muscle.

It is concluded that transmission of inhibition from the plantar nerve to soleus muscle is modulated during walking. It would minimize this inhibition during the stance phase of walking and might enhance soleus muscle activity via this reflex pathway for the support of weight.
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Theme: Motor systems and sensorimotor integration
Topic: Spinal cord and brainstem

Keywords: Plantar nerve; Soleus; Ib inhibition; Walking; Weight loading; Modulation

1. Introduction

Hindlimb extensor group I muscle afferents evoke inhibitory reflexes at oligosynaptic latencies in cat extensor motoneurones [15]. This non-reciprocal group I inhibition is considered a feedback system whereby proprioceptive activities in tendon organ, and primary muscle spindle afferents inhibit extensor motoneurones via a reflex pathway [14]. During both fictive [7,11,18] and overground [13] locomotion, however, stimulation of the same afferents evokes extensor excitation [18,22]. Thus, extensor group I reflexes result in a proprioceptive feedback system that can increase extensor activity and provide additional weight support in both cats [13,18] and humans [12,22]. Based on these findings, during locomotion the Ib inhibitory pathway closes and Ib extensor facilitatory pathway opens. Plantar stimulation enhanced the ongoing activity in medial gastrocnemius motoneurones during periods of rhythmic locomotor bursts in clonidine-treated spinal cats [19].

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doi:10.1016/j.brainres.2005.04.043

Synaptic connections to human leg and thigh muscles from intrinsic plantar muscles have been reported [17]. They investigated group I projections from intrinsic foot muscles to motoneurones of the leg and thigh muscles in humans based on the post stimulus time histogram (PSTH) and H-reflex. They showed that there was weak heteronymous monosynaptic Ia excitation and later inhibition in the extensor muscles. They speculated that this later inhibition is disynaptic Ib inhibition because of its longer latency than monosynaptic excitation, longer duration, and the low threshold of the effect and of its sharp onset.

In the present study, we first determined that short latency inhibition from the plantar nerve is attributable to Ib inhibition, then studied modulation of this inhibition in the stance phase of walking and during voluntary tonic and dynamic plantar flexion. Moreover we investigated the influence of body unloading on this inhibition. Taken together, our findings indicate that the inhibition from the plantar muscle of the foot to the soleus muscle is decreased during the stance phase of walking.

2. Materials and methods

2.1. Subjects

The subjects in experiment 1 were 5 healthy men aged 24–46 years. Each experiment was repeated 5 times on 2 subjects. In experiments 2 and 3, the subjects were 5 healthy men aged 26–44 years. In all experiments, everyone gave his informed consent to the experimental procedures, which had been approved by the local ethics committee.

2.2. Experiments

2.2.1. Experiment 1

The subject sat in a reclining armchair with the right foot attached to a footplate connected to a torque meter. The torque exerted on the footplate and the voluntary rectified and integrated electromyographic activity (EMG) measured by surface electrodes placed on the soleus muscle were displayed on an oscilloscope placed in front of the subject. The hip was semiflexed 120° and the knee flexed 160°. Plantar flexion of the ankle was 110°.

Surface electrodes were used both for stimulation and recording. Excitability of the soleus motoneuron was estimated by means of the H-reflex and the rectified averaged surface EMG. Paired recording electrodes (10 mm in diameter) were placed 2.0 cm apart over the distal part of the soleus muscle and connected to conventional amplifiers (frequency band, 50-3 kHz). The soleus H-reflex was evoked by stimulating the posterior tibial nerve (PTN) with isolated constant current through a monopolar electrode (1 ms rectangular pulse) placed in the popliteal fossa (Nihon Kohden, electric stimulator and isolator, Nihon Kohden, SS-104 J). The indifferent electrode was placed below the patella. At the beginning the maximal M response (M_{max}) was measured as the peak to peak value. The H-reflex was also measured as the peak to peak value, expressed in relation to M_{max} (%M_{max}). The size of the control H-reflex during the experiment was adjusted to give an H-reflex of 15–25% of M_{max} at rest and during walking (see below), because sensitivity of the H-reflex to the conditioning input depends on reflex size [9].

The conditioning stimulus (rectangular 1 ms pulse) was delivered to the branch of the tibial (plantar) nerve in the internal malleolus of the ankle joint. Intensities of the conditioning stimuli were expressed as multiples of the threshold for the direct M response (×MT) of the flexor digitorum brevis (FDB) muscle. To investigate the circuit of inhibition from the plantar nerve, we examined the convergence of this inhibition with other segmental inhibitory pathways for disynaptic reciprocal Ia inhibition, Ib inhibition, and inhibition from the cutaneous nerve. Conditioning stimuli were applied to the common peroneal nerve (CPN) at the level of the caput fibulae and an optimal conditioning test interval of about 2 ms to test disynaptic reciprocal Ia inhibitory pathway activity [8,23]. Ib inhibition is induced by a conditioned stimulus to the medial gastrocnemius (MG) nerve [20]. To test for inhibition from the cutaneous nerve, the tip of the big toe was conditioned by triple shocks (0.5 ms pulse, 250 Hz) through a surface electrode. Stimuli were given every 3 s in the H-reflex study in a randomized, conditioned, and unconditioned test stimuli sequence, with randomization of the conditioning test interval. At least 10 reflexes (conditioned and unconditioned) were measured in each series for later statistical analysis (mean, standard error of the mean, and differences between groups by the Student's paired t test).

In the averaged EMG study, the conditioning stimulus was given every 1 s to the plantar nerve while the subject performed tonic plantar flexion. The rectified EMG triggered by the stimulus was averaged over the period from 50 ms prior to the beginning of stimulus to 150 ms after, usually 100–200 sweeps (Nihon Kohden, Medical Data Processor, ATAC 3300). The background EMG was measured during the 50-ms period preceding stimulus.

2.2.2. Experiment 2

The purpose was to evaluate the modulation of inhibition from the plantar nerve to the soleus muscle during the stance phase of walking. This experiment also used the H-reflex and averaged surface EMG. Stimuli and recordings were the same as in Experiment 1. The conditioning intensity of the plantar nerve was 1.2 ×MT in the FDB muscle. Checks were made throughout the experiments by monitoring the size of the small M response in the FDB muscles that no systematic changes occurred in the position of the stimulat-

ing electrode. All trials in which the size of the average *M* response deviated more than 10% from the pre-set level were excluded from further analysis.

The individuals' tasks were walking on a treadmill, and tonic and dynamic plantar flexion in the sitting position.

2.2.2.1. Tonic plantar flexion. Each subject was requested to maintain a steady torque and EMG activity level in the soleus muscle at either 4 or 5 prescribed levels of contraction ranging from 5% to 40–50% of the maximal voluntary plantar flexion force. Stimulation of the plantar nerve produced depression of the stimulus-triggered average and rectified soleus EMG. The amount of inhibition was the difference between the level of EMG activity from the onset of inhibition and that of the background EMG activity. During tonic contraction, the background EMG activity determined was the mean EMG activity level 50 to 5 ms before stimulus. We also measured inhibition during standing before walking. Measurements during tonic contraction were made at the beginning of the experiment and repeated after the walking session.

2.2.2.2. Walking. Treadmill speed was set at 4 km/h. A pressure-sensitive trigger placed under the heel of the subject's right shoe was used to trigger a stimulus which could be delivered at different intervals in relation to the heel strike during walking. In all the situations, the background EMG activity were amplified, bandpass filtered (20 Hz to 1 kHz), rectified on a Mac Lab (ADInstruments, USA), and sampled (5 kHz sampling frequency) in a Mac Powerbook (Apple, USA). Care was taken that measurements were made at matched levels of background EMG activity in the various situations.

Differences between the observed inhibition and the expected inhibition were tested for statistical significance with Wiscoxon's signed rank test and a two-way ANOVA. The level of significance was set to 0.05.

2.2.2.3. Dynamic plantar flexion. The rectified, integrated soleus EMG activity (average of 100 walking cycles) measured during the stance phase of walking was displayed on an oscilloscope in front of the subject. The task was to match both the level of and rate of change in soleus EMG activity by performing isometric dynamic plantar flexion. Most subjects had considerable difficulty achieving an acceptable match between EMG activities in the two tasks, generally only being achieved after several repetitions of the task. Measurements were made at various intervals in relation to the onset of EMG activity. Measurements during dynamic contraction were obtained after the walking session.

2.2.2.4. H-reflex study during walking. Because the H-reflex is strongly modulated during walking [5], the control H-reflex size could be obtained only by adjusting the intensity of PTN stimulation. Since the intensity of the PTN stimulation, which evoked the H reflex, was generally below the *M* threshold, this could be done by alternating the stimuli which evoked the H reflex with stimuli at a slightly higher intensity to evoke an *M* response. Throughout the experiments checks were made by monitoring the size of the *M* response in the soleus muscle after adjusting the intensity that no systematic changes had occurred in the position of the stimulating electrode. All trials in which the size of the average *M* response deviated more than 10% from the pre-set level were excluded from further analysis.

Conditioning stimulation intensity was the same for the averaged EMG. At the beginning of each experiment the time course of the effect of plantar nerve stimulation on the H-reflex was obtained while the subject sat at rest (Fig. 1B). The interval in which the maximal amount of inhibition occurred was used for the rest of the experiment. Maximal inhibition usually was produced at the conditioning test interval of 10–12 ms in the various subjects. Control and conditioned reflexes were alternated randomly at a PTN stimulation of higher intensity than that used to evoke the H-reflex (see above). Inhibition of the conditioned reflex was expressed as a percentage of the size of the control H-reflex size. In each situation, at least 100 control and conditioned reflexes were averaged for each point. The statistical significance of changes in the H reflex was tested by the Student's paired *t* test.

2.3. Experiment 3

The purpose was to evaluate the influence of unloading to the modulation of the inhibition from the plantar nerve during the stance phase of walking by averaged surface EMG. Stimuli and recordings were the same as in Experiment 2. Moreover, we examined the effect of unloading of other segmental inhibitory pathways for disynaptic reciprocal Ia inhibition, Ib inhibition. Conditioning stimuli were applied to the CPN at the level of the caput fibulae for investigating Ia inhibition. Ib inhibition is induced by a conditioned stimulus to the MG nerve. The conditioning intensity of the CPN and MG nerve was also 1.2 ×MT in the TA muscle and MG muscle, respectively. Checks were made by monitoring the size of the small *M* response in the each muscle as same in Experiment 2.

First, each subject was requested to maintain a steady torque and EMG activity level in the soleus muscle for investigating the relationship between the inhibition and EMG activity as in Experiment 2 by stimulating each nerve.

Second, the subjects were walking with body unloading 0%, 30%, and 60%. Unloading was achieved by suspending the subjects from a parachute harness connected to an overhead crane. The degree of unloading was provided by a counterweight and was varied randomly for each subject. We compared the amount of inhibition by plantar nerve

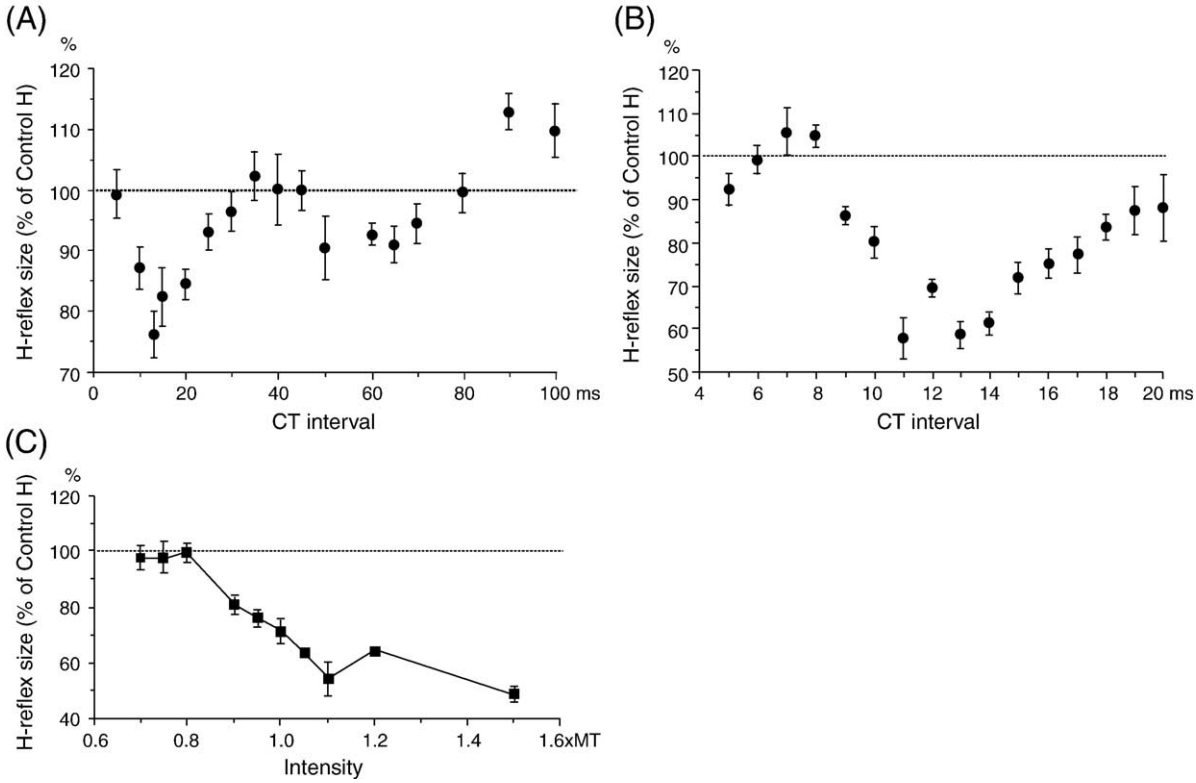

Fig. 1. Changes of H-reflex of soleus muscle by conditioning stimulation of the plantar nerve. (A and B) Time courses of effects of the soleus H-reflex evoked by a single conditioning stimulus (intensity 1.0 ×MT, duration 1 ms) to the plantar nerve at rest. The abscissa shows conditioning stimulus intervals in ms, the ordinate the size of the conditioned reflex, expressed as a percentage of the unconditioned value. Each bar represents one standard error of the mean. (C) Relation between inhibition and the strength of the conditioning stimulation at the conditioning test interval of 13 ms. The abscissa shows the strength of the conditioning stimulus expressed as multiples of the threshold for the direct *M* response (×MT) of the flexor digitorum brevis (FDB) muscle.

stimulation, CPN stimulation, and MG stimulation during unloading 0%, 30%, and 60%.

3. Results

3.1. Effects from the plantar nerve to soleus muscle

Fig. 1A shows the time course of the effects of the soleus H-reflex evoked by a conditioning stimulus to the plantar nerve at rest. Clear inhibition occurred at about 15 ms with a return to the control level at about 40 ms; thereafter, there was weak inhibition at about 60–70 ms. To measure the conduction velocity of the afferent fibers, the tibial nerve was stimulated in the popliteal fossa and the internal malleolus of the ankle joint, and the H-reflex was evoked in the FDB muscle. Then conduction velocity was calculated from the distance between the electrodes and the differences in the latencies of H-reflexes. In this individual, the distance between the conditioning and test stimulation positions was 40 cm, and the conduction velocity of the afferent fibers which evoked the H-reflex in the FDB muscle was 50.6 m/s.

Fig. 1B shows the short time course within 20 ms. Inhibition started at 9 ms, but just before that weak facilitation appeared at 7–8 ms. Short latency inhibition was present in all the subjects, but the facilitation just before inhibition was not very clear and not always present. Early inhibition reached a peak 4–5 ms after onset and continued for more than 10 ms. Fig. 1C shows the inhibition at the conditioning test interval 13 ms when the conditioning stimulus strength was changed systematically. It occurred at the conditioning stimulus strength between 0.8 and 0.9 ×MT, and almost reached a plateau above 1.1 ×MT. Inhibition therefore was clearly evoked below the threshold of the motor fibers.

Fig. 2 shows the effects of stimulation of the plantar nerve on the averaged and rectified FDB and soleus surface EMGs when the subject performed tonic cocontraction of both muscles. When the strength was 0.79 ×MT, no significant changes were seen. At the strength of 0.83 ×MT, a slight increase occurred in the EMG at 50 ms only in the FDB muscle (white arrow), which was attributed to a monosynaptic reflex on its latency and the conditioning strength. When the strength was increased to 0.91 ×MT, the increase in the EMG of the FDB muscle was enhanced, whereas there was a decrease in the EMG for the soleus muscle (black arrow). When the strength was increased to more than 0.91 ×MT, the increase in the EMG of the FDB muscle and the decrease in the soleus muscle were equally enhanced (Figs. 2D and E). On the basis of the averaged EMG findings,

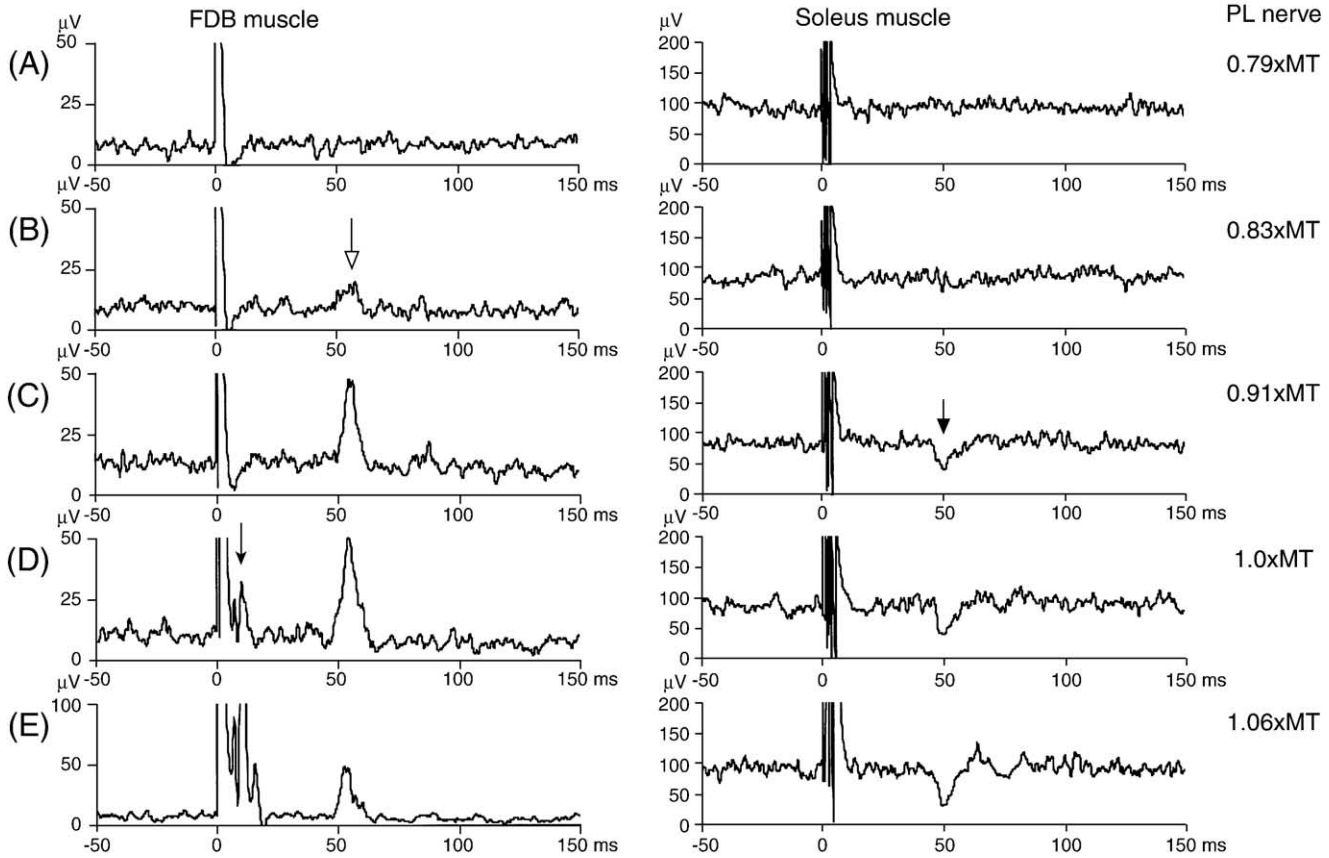

Fig. 2. Changes in averaged surface EMGs of FDB and soleus muscles due to conditioning stimulation to the plantar nerve. Averaged surface FDB (left side) and soleus (right side) EMGs triggered by a conditioning stimulus to the plantar nerve during tonic contraction of the muscles for 100 sweeps when the conditioning stimulus strength was changed from 0.79 to 1.06 ×MT. The ordinate shows amplitudes of the rectified EMGs (microvolts), the abscissa the latency after conditioning stimulation to the plantar nerve.

the threshold at which the afferent fibers produced inhibition was lower than that of the motor fibers.

Convergences were investigated of inhibition from the plantar nerve with other segmental inhibitory pathways: disynaptic reciprocal Ia inhibition, Ib inhibition, and inhibition from the cutaneous nerve. Ia inhibition was evoked by a conditioning test interval of 2–3 ms at the stimulus of 0.85 to 0.90 ×MT to the CPN. Ib inhibition was evoked by a conditioning test interval of 5 ms at 0.90 to 1.0 ×MT to the MG nerve. Inhibition from the cutaneous nerve was evoked by a conditioning test interval of 20–25 ms at 1.5 ×PT (perceptional threshold); painless stimulation to the tip of the big toe through the surface electrode. The plantar nerve was stimulated at the conditioning test interval of 10–12 ms and a strength of 0.7 to 0.85 ×MT. Fig. 3 shows results of the pooled data for 5 subjects. Ia inhibition and inhibition from the cutaneous nerve did not converge when there was inhibition from the plantar nerve (Figs. 3A and C). In contrast, Fig. 3B shows the inhibition produced by plantar nerve stimulation (PL), Ib inhibition from MG nerve stimulation (MG), algebraic sum of individual stimulation, and combined stimulation (PL + MG). The inhibition from combined stimulation had a statistically different significance that differed from that of algebraic sum ($P < 0.05$).

3.2. Modulation of plantar nerve-induced inhibition of soleus background EMG activity during walking

Fig. 4 shows the relationship between the amount of inhibition and the background EMG. Fig. 4A shows the stimulus-triggered averaged soleus EMG when the test force was 5%, 20%, 30%, and 40% of the maximal voluntary contraction force. Inhibition increased with the increase in the background EMG. Fig. 4B shows the linear relationship between the background soleus EMG and inhibition from the plantar nerve to the soleus muscle. The same amount of inhibition occurred between tonic contraction at the same background EMG level while sitting or standing. In all the experiments, regression lines were fitted to the data and used to calculate the amount of inhibition to be expected during walking at a given level of background EMG activity.

Fig. 5A shows inhibition from the plantar nerve to the soleus muscle during walking on a treadmill as determined by the averaged surface EMG. Background soleus EMG

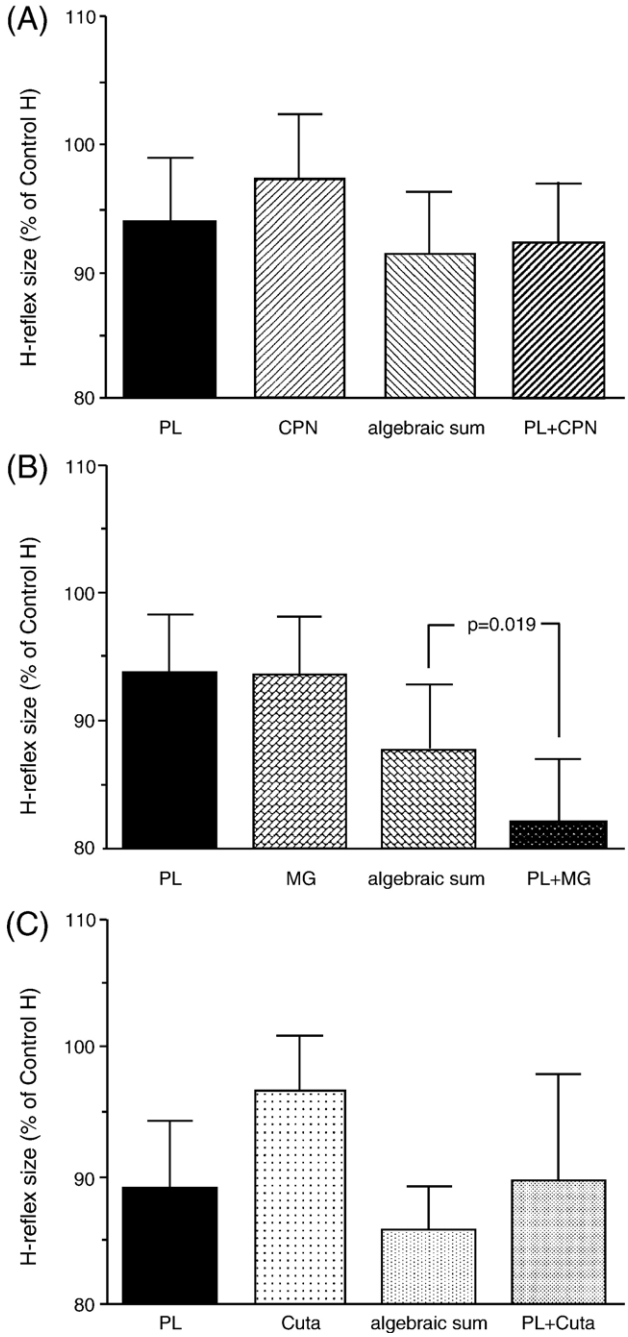

Fig. 3. Convergence of inhibitions from the plantar nerve and other segmental inhibitory pathways. Panels A, B, and C show pooled data for the five subjects. (A) Plantar nerve stimulation (PL), common peroneal nerve stimulation (CPN), algebraic sum, both nerves stimulated (PL + CPN). (B) Plantar nerve stimulation (PL), medial gastrocnemius nerve stimulation (MG), algebraic sum, both nerves stimulated (PL + MG). (C) Plantar nerve stimulation (PL), cutaneous nerve stimulation (Cuta), algebraic sum, both nerves stimulated (PL + Cuta). Each bar represents the mean and one standard deviation. *Statistically significant difference ($P < 0.05$).

activity usually increased steadily throughout the stance phase (bottom Fig. 5A). Although inhibition from plantar nerve stimulation increased with the increase in background soleus muscle activity during tonic plantar flexion (Fig. 4B), but during the stance phase this did not occur. Fig. 5B clearly shows the differences between the amount of inhibition in the stance phase of walking and during tonic plantar flexion in pooled data for the 5 subjects. Clearly, in the early stance phase there was less inhibition than expected from the amount of inhibition measured during tonic plantar flexion. Toward the end of the stance phase the difference is more evident. The signed rank test done on the group data shows a clear difference between the observed and expected inhibition during the stance phase, the difference being clearer toward the end of stance and reaching statistical significance (signed rank test, $P < 0.05$).

3.3. Dynamic plantar flexion

During walking, background EMG activity steadily changes, whereas it is almost constant during tonic contraction. To investigate whether this difference in background EMG activity explains the observed difference

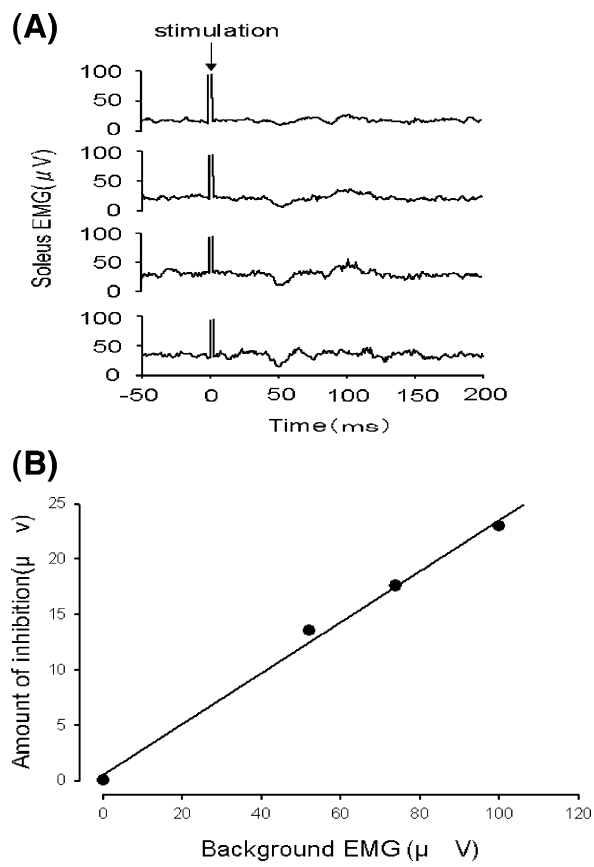

Fig. 4. Inhibition during tonic plantar flexion at various force levels. (A) Traces show the depression of rectified soleus EMG activity during tonic voluntary plantar flexion at various forces that were 5%, 20%, 30%, and 40% of the maximal voluntary contraction force from the upper trace. Sweeps were triggered by the respective stimuli (100 sweeps averaged). The latency of inhibition was about 45 ms. (B) The amount of inhibition (difference in microvolts between the amount of background EMG activity and EMG activity during inhibition) increases almost linearly with the amount of background EMG activity.

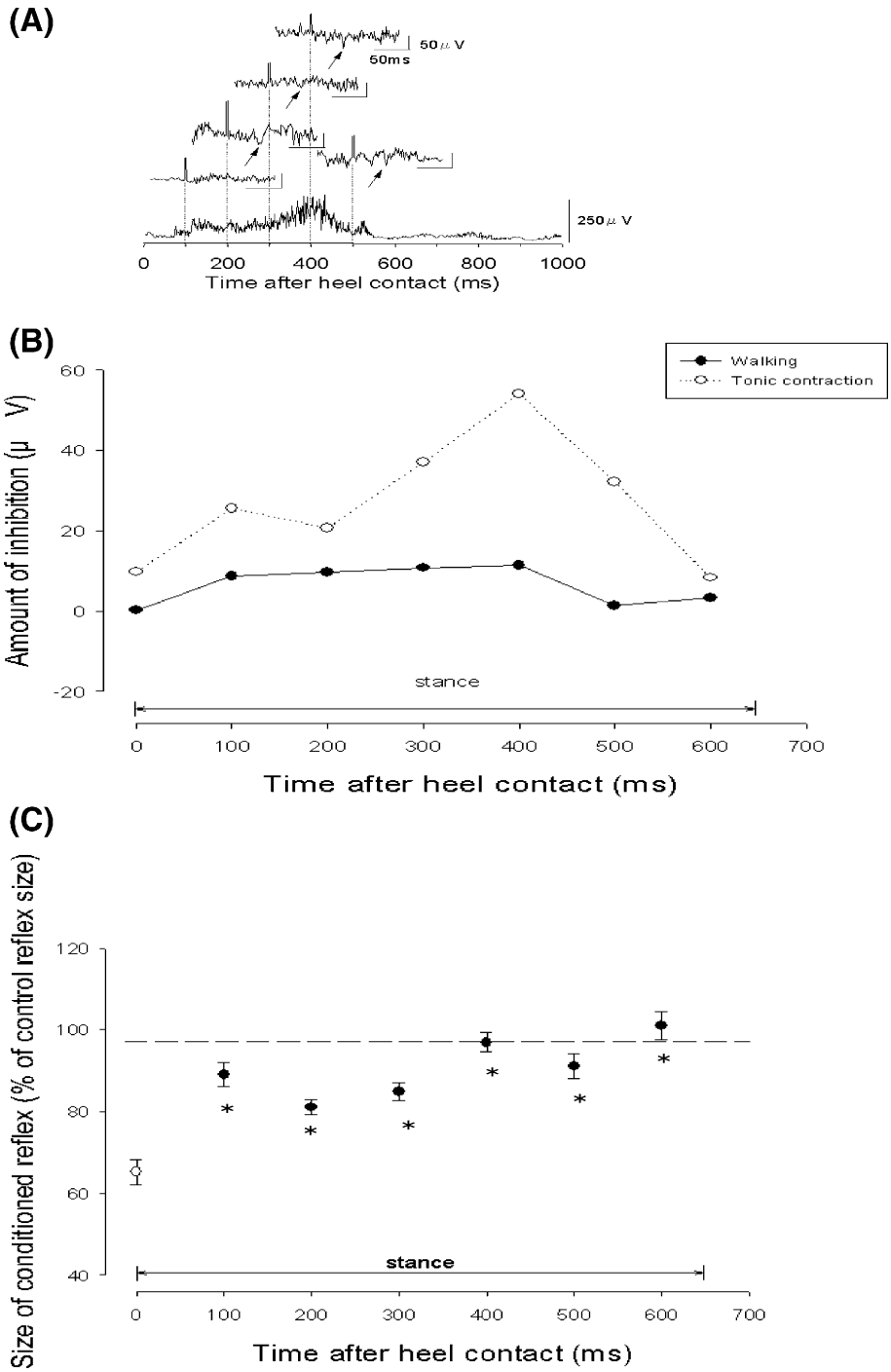

Fig. 5. Modulation of inhibition of the averaged surface soleus EMG activity and H-reflex during walking. (A) Inhibition of soleus EMG activity induced by stimulation of the plantar nerve (1.2 ×MT) during the stance phase of walking. Arrows indicate the approximate onset of inhibition. Traces show the stimulus-triggered averaged (100 sweeps) and rectified soleus EMG activities at various times after heel strike. The lower part shows soleus EMG activity (average of 100 sweeps) during the stance phase, the abscissa, and the interval after heel contact. (B) Pooled data for all the subjects about modulation of inhibition of the averaged soleus EMG activity. The amount of inhibition (closed circles) during walking (traces in A) together with the expected inhibition (open circles), calculated as above. (C) Pooled data for all the subjects about the modulation of plantar nerve stimulation on the soleus H-reflex in the stance phase of walking. The conditioning test interval was 12.0 ms for all the subjects. Symbols represent averages of 100 conditioned and 100 control responses. Each bar represents one standard error of the mean. The open circle at the left in each the graph gives the size of the conditioned reflex as a percentage of the control H-reflex when the subjects were at rest. $^*P < 0.05$ (Student's t test).

in inhibition during walking and tonic contraction, we investigated the amount of inhibition during dynamic plantar flexion, in which both the levels of and rate of change in EMG activity were matched to values obtained in the stance phase of walking on all the subjects. Plantar nerve stimulation evoked almost the same inhibition of soleus

EMG activity during dynamic plantar flexion as expected from tonic contraction irrespective of the time after the onset of movement or EMG activity (no figures). Inhibition was less in everyone during walking than during dynamic plantar flexion at matched levels of background soleus EMG activity.

3.4. Modulation of inhibition of the soleus H-reflex evoked by plantar nerve stimulation during walking

The time course of the effect of plantar nerve stimulation on the soleus H-reflex was investigated in all five subjects at the beginning of the experiment, and the interval of maximal inhibition was used in the rest of the experiment. Fig. 5C shows the size of the conditioned reflex as a percentage of the control H-reflex size during this conditioning test interval in the five subjects. Plantar stimulation evoked less inhibition of the H-reflex in all of the subjects during that phase, significant inhibition being evoked in everyone at rest. The signed rank test used on the group data showed a clear difference between the observed and expected inhibition during the stance phase, which difference was more clear toward the end of stance and became statistically significant (signed rank test, $P < 0.05$).

3.5. Effect of unloading for the spinal segmental inhibitory pathway

Same as Fig. 4, we investigated the relationship between the amount of inhibition and the background EMG. We observed the linear relationship between the background soleus EMG and inhibition from the plantar nerve to the soleus muscle, inhibition from the CPN, inhibition from the MG nerve, respectively. The same amount of inhibition occurred between tonic contraction at the same background EMG level while sitting or standing.

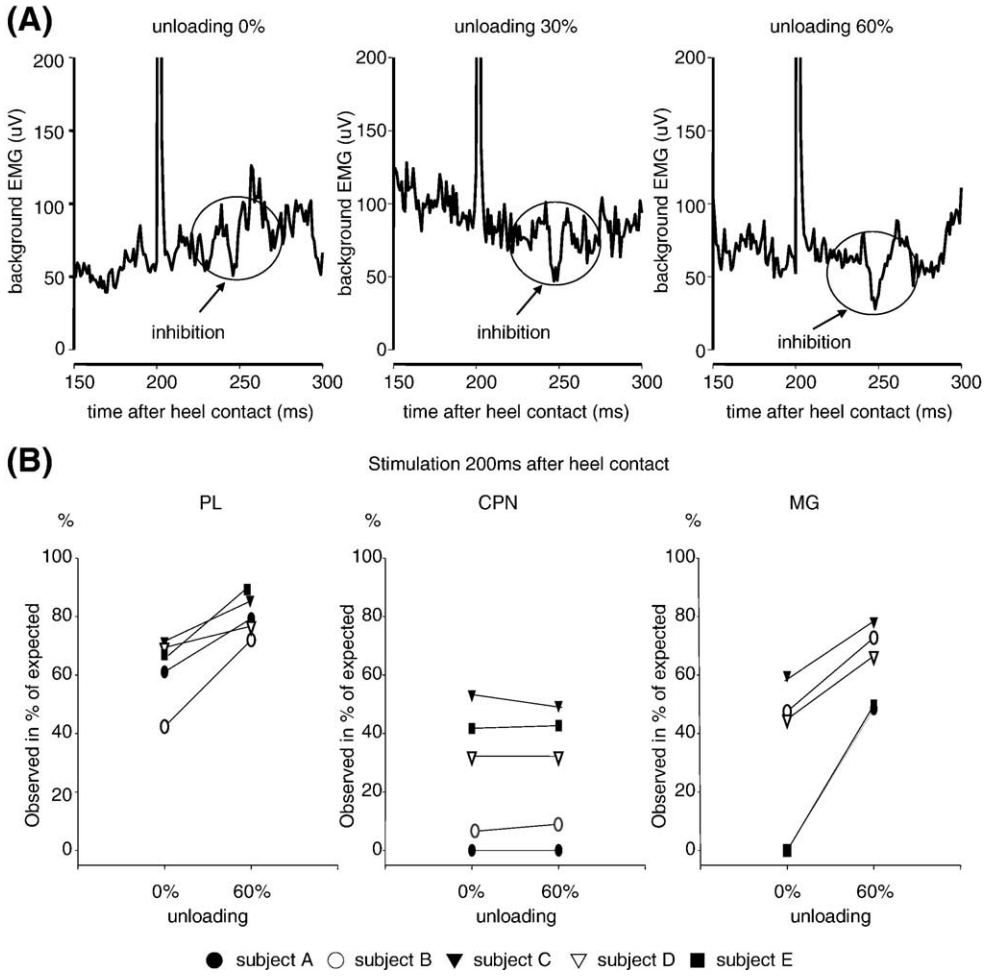

Fig. 6. Effect of body unloading to the modulation of inhibition of the averaged surface soleus EMG activity during walking. (A) From the left, the effect of 0%, 30%, and 60% unloading on the inhibition of EMG activity of the soleus muscle. The amount of inhibition to the soleus muscle from the plantar nerve increased, while unloading increased. Traces show the stimulus-triggered averaged (100 sweeps) and rectified soleus EMG activities at 200 ms after heel strike. (B) These figures show the amount of inhibition/expected inhibition induce by plantar nerve stimulation (PL), common peroneal nerve stimulation (CPN), and medial gastrocnemius nerve stimulation (MG) during unloading 0% and 60% for each subject. The inhibition from the plantar nerve and MG nerve increased by unloading 0% to 60%, while the amount of the disynaptic Ia inhibition did not change in each subject.

We observed the soleus EMG activity slightly reduced during unloading 30% and 60%. However, there was no significant difference in the duration of the stance phase between the unloading conditions. Fig. 6A shows the inhibition from the plantar nerve to the soleus muscle during walking with body unloading 0%, 30%, 60% on a treadmill as determined by the averaged surface EMG method. Inhibition of soleus EMG activity was induced by stimulation of the plantar nerve (1.2 ×MT) during the stance phase of walking. The amount of inhibition increased while unloading increased.

As mentioned above, we compared the amount of inhibition corrected for background soleus EMG activity between the unloading conditions. Fig. 6B showed the amount of inhibition between unloading 0% and 60% during walking. These figures show that inhibition from the plantar nerve stimulation and inhibition from MG nerve increased by unloading, but the inhibition from CPN stimulation to the soleus EMG did not change.

4. Discussion

Our present findings can be summarized as follows: (1) short latency inhibition on the soleus motoneuron was evoked by low strength stimulus of the plantar nerve in the internal malleolus of the ankle joint, and this inhibition converged to Ib inhibition from the MG nerve, not to disynaptic Ia inhibition or inhibition from the cutaneous nerve. (2) Short latency inhibition was modulated during walking, decreasing toward the end of the stance phase. (3) The amount of short latency inhibition from the plantar nerve and Ib inhibition from the MG nerve on the soleus motoneuron during walking was increased by unloading of body weight.

4.1. Short latency inhibition from the plantar nerve to soleus muscle

The reflex pattern for the soleus EMG produced by stimulation of the branch of the tibial nerve (plantar nerve) begins with inhibition and is followed by facilitation in the cat [1] and humans [3].

In this study short latency inhibition was evoked at a strength below 1.0 ×MT and started from 9 ms at conditioning test interval. The conduction velocity of the afferent fibers in which the H reflex was evoked in the FDB muscle was 50.6 m/s. Taking into account the latency and distance of the conditioning test electrodes, the inhibition may have been due to oligosynaptic inhibition from the plantar nerve. Based on the averaged EMG (Fig. 2), the threshold of the afferent fibers is slightly higher than that of the Ia fibers and lower than that of the motor fibers. The inhibitory pathway from the plantar nerve did not converge to disynaptic Ia inhibition and inhibition from the digital nerve but to Ib inhibition from the MG nerve. This short latency inhibition from the plantar nerve therefore is considered to include Ib inhibitory interneurones, which supports the hypothesis of Marque et al. [17].

The responses produced most likely were of muscular origin. Although we found no convergence of the cutaneous afferents to the inhibition from the plantar nerve to the soleus muscle, cutaneous afferents contribution to inhibition from the plantar nerve stimulation cannot be ruled out. Pierrot-Deseilligny et al. [21] reported that sural nerve stimulation depresses the Ib reflex pathway in the quadriceps and biceps femoris muscles, not the soleus muscle. Yang and Stein [25] stimulated the tibial nerve at the ankle, as we did in this report. They found responses at a middle latency between 50 and 90 ms when the tibial nerve was stimulated 1.5–2.0 times the motor threshold. They concluded that the responses most likely were of cutaneous origin because they closely resembled responses to stimulation of the sural nerve. However, the inhibition found in our study seems to differ from that of the cutaneous reflex with respect to the intensity of the conditioning stimulus and latency of the responses.

4.2. Modulation of inhibition from the plantar nerve to soleus muscle in the stance phase of walking

In our study, based on the averaged surface EMG and the H-reflex inhibition in the soleus muscle after stimulation of the plantar nerve decreased toward the end of the stance phase. Selective activation of Ib afferents by electrical stimulation of plantar afferents has been reported because Ib rather than Ia afferents from the plantar nerves are associated with MG motoneurones in cats [19]. Weak stimulation of the plantar nerve therefore provides reliable activation of Ib afferents. In spinal cats, stimulation of the plantar nerve prolonged extensor bursts and delayed the appearance of flexor bursts in rhythmic locomotor neural activity after an injection of L-DOPA [7]. Similarly, plantar stimulation enhanced ongoing activity in MG motoneurones during periods of rhythmic locomotor bursts in clonidine-treated spinal cats [19]. Therefore, the group Ib input resulted in clear excitation during the extensor phase of the fictive step cycle in cats.

Stephens and Yang [22] investigated the behavior of the putative group Ib reflex in human extensor muscles of the ankle during sitting and walking. They found that the inhibition produced by afferents from the medial gastrocnemius nerve in the soleus muscle decreased during walking and that the change from sitting to walking was small. They speculated that this small change was due to giving an artificial, electrical stimulus to one muscle nerve. We, however, think there is another possibility, that the stimulating electrode to the MG nerve moves during walking, but it is difficult to monitor the status of conditioning stimulation because they stimulated MG nerve at subthreshold of motor nerve.

Although short latency excitation during the stance phase occurred in some of our and Stephens and Yang [22] subjects, as a group the effect was reduced inhibition. In the cat experiments, the extensive removal of afferent inputs might have rendered the central pattern generator more susceptible to the remaining input. In normal humans, the influence of the higher center overshadows these effects. Bipedal and quadrupedal locomotion also may have effects.

Yang and Stein [25] found reflex reversal of the cutaneous reflex during walking. Cutaneous responses were facilitatory in the soleus muscle at the beginning of the stance phase but became inhibitory at the end of that phase. The cutaneous reflex latency ranged from 50 to 90 ms, slightly longer than in our study. Moreover, in our study inhibition after stimulation of the plantar nerve decreased toward the end of the stance phase. Christensen et al. [6] reported that the cutaneous reflex in human tibialis anterior muscle after stimulation of the sural nerve is modulated during walking. Possibly, cutaneous reflexes from the tibial nerve area of the foot to the leg muscles are modulated during walking, we think that the cutaneous contribution to the inhibition was low in our study because the response differs markedly and is modulated differently during walking. In the preliminary experiment, we conducted the similar experiment by stimulating the big toe during walking; however, we did not find the modulation during the stance phase.

4.3. Effect of unloading to the spinal inhibitory pathway

In the cat experiments, homonymous force feedback from Golgi tendon organs has a substantial contribution to ongoing ankle extensor activity during walking in the cat. Group Ib pathways from ankle extensors have an excitatory action on extensor activity during walking in spinal and decerebrate cats [19,24] as well as fictive locomotion in spinal and decerebrate cats [2,7,10]. Therefore, our results confirm and support that the Ib pathways in human might also be the load receptor during walking, because the inhibition from the plantar nerve and the Ib inhibition from the MG nerve were increased by unloading during walking. Ia inhibition from the CPN nerve did not change, so unloading may not influence Ia inhibitory pathway.

4.4. Functional significance

In the stance phase, the foot muscle usually contracts to maintain arching the arch of the foot but is stretched gradually during that phase because of weight bearing. We concluded that foot muscle contraction might support soleus muscle activity during the stance phase.

Intrinsic plantar muscles are especially active at the end of the stance phase in human walking [16] and running, showing lengthening contraction that evokes strong muscle spindle afferent discharges [4]. The functional role of excitatory Ia projections must be considered, whether or not transmission of the opposite group I inhibitory effects on extensor motoneurones is assumed to be depressed. Intrinsic plantar muscle contraction therefore also may support soleus muscle contraction via non-monosynaptic Ib inhibition for support of the load during the stance phase.

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